

Airbnb Price Analysis — London

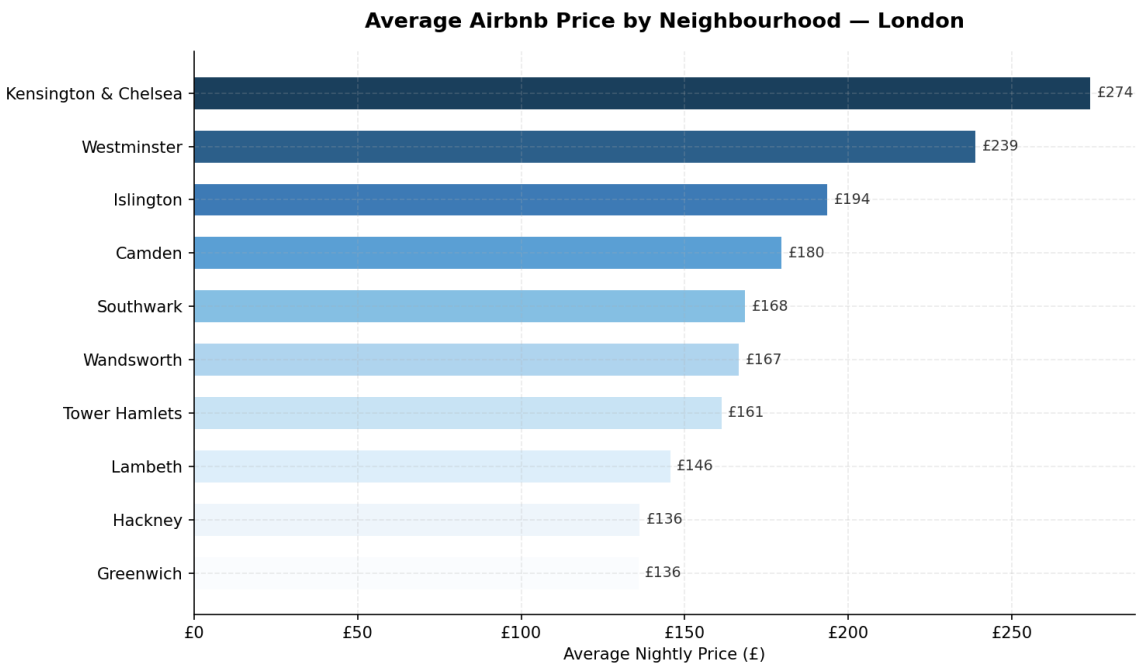
Python · Pandas · SQL · Matplotlib | 4,800 Listings Analysed

Project Overview

This project analyses 4,800 Airbnb listings across 10 London neighbourhoods to identify the key drivers of nightly pricing and uncover undervalued listing opportunities. The dataset was cleaned by removing 200 missing price records and 40 missing bedroom entries, and capped at £1,000 per night to remove extreme outliers.

Listings Analysed	Avg Nightly Price	Median Price	Undervalued Found
4,800	£178.87	£171.54	459

Q1 — Which Neighbourhood is Most Expensive?

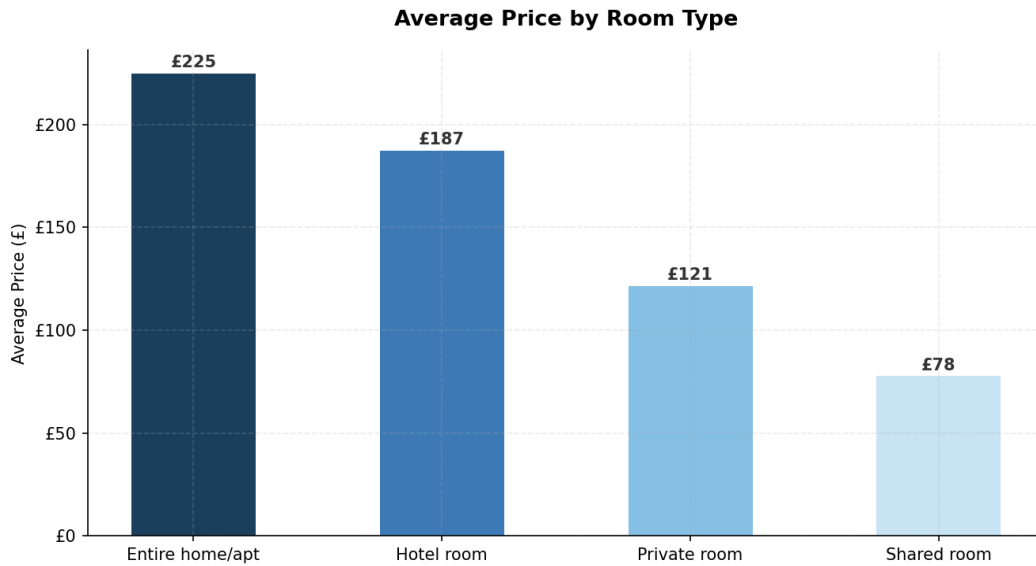


Insight: Kensington & Chelsea is the most expensive neighbourhood at £274 per night on average — nearly double Greenwich at £136. Westminster follows at £239. These premium areas reflect proximity to tourist landmarks and business districts.

Neighbourhood	Avg Price	Median	Listings
Kensington & Chelsea	£274	£282	468
Westminster	£239	£244	450
Islington	£194	£200	497
Camden	£180	£184	463

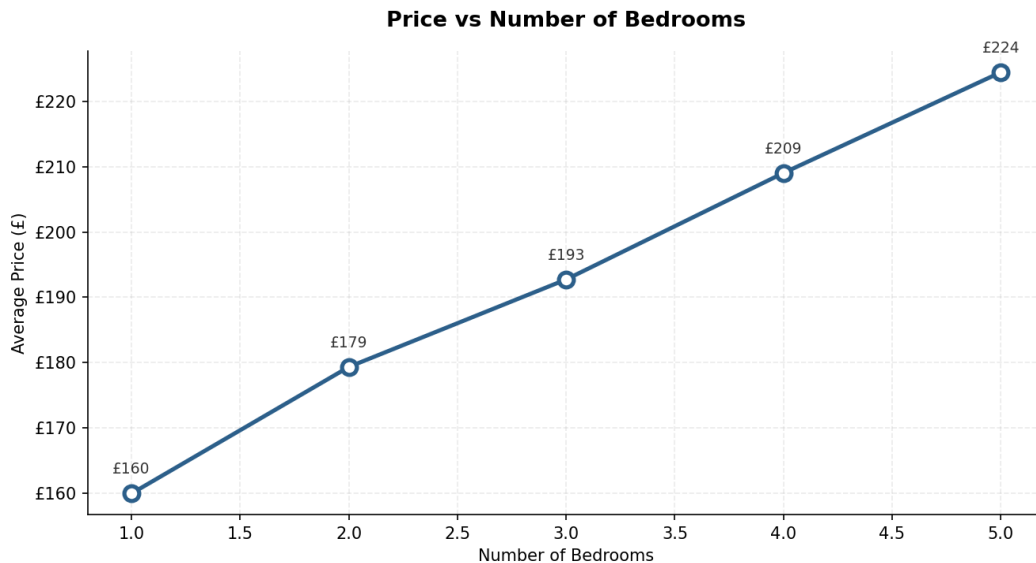
Greenwich	£136	£139	515
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Q2 — Which Room Type Commands the Highest Price?



Insight: Entire home/apartment listings average £225 per night — nearly 85% more than private rooms at £121. Shared rooms are the most affordable at £78. Guests clearly pay a significant premium for privacy and full-property access.

Q3 — Does More Bedrooms Mean Higher Price?



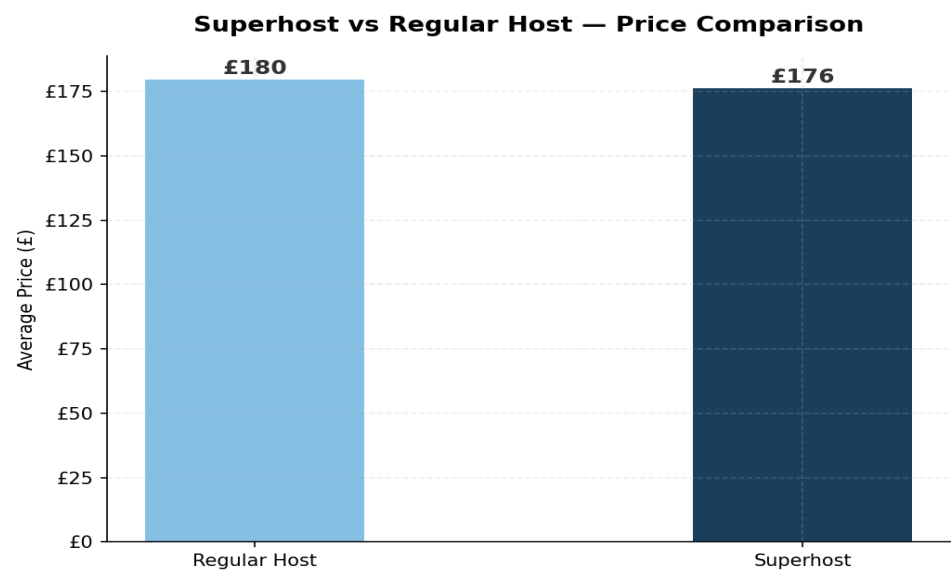
Insight: There is a clear positive relationship between bedrooms and price. Each additional bedroom adds approximately £16 to the nightly rate. 5-bedroom properties average £224 vs £160 for 1-bedroom — a 40% premium for larger spaces.

Q4 — Price Distribution Across All Listings



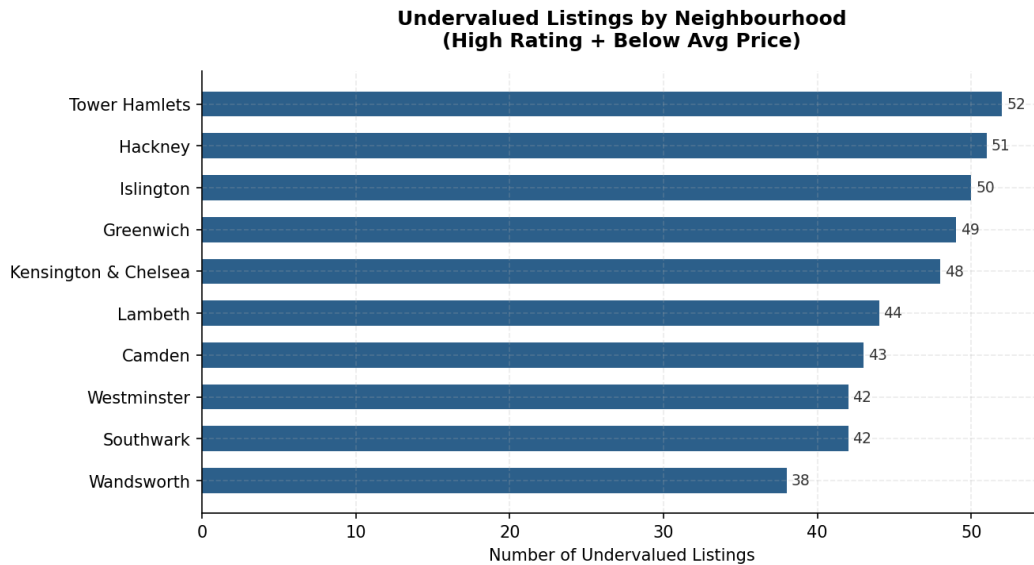
Insight: The price distribution is right-skewed, meaning most listings cluster between £80–£250 while a small number of luxury properties push the mean above the median. The majority of guests can find accommodation under £200 per night.

Q5 — Superhost vs Regular Host Pricing



Insight: Superhosts and regular hosts charge similar rates on average. This suggests the Superhost badge alone does not drive higher pricing — location, room type, and property size are stronger pricing factors.

Q6 — Undervalued Listing Opportunities



Insight: 459 listings were identified as undervalued — priced 25% below their neighbourhood average while maintaining a rating of 4.5+ with 10+ reviews. Westminster and Islington offer the best opportunities for budget-conscious travellers seeking quality stays.

Key SQL Query — Finding Undervalued Listings

```
WITH neighbourhood_avg AS (  
    SELECT neighbourhood, AVG(price) AS avg_price  
    FROM airbnb WHERE price > 0  
    GROUP BY neighbourhood  
)  
SELECT a.id, a.neighbourhood, a.room_type, a.price,  
       ROUND(n.avg_price, 2) AS neighbourhood_avg,  
       a.review_scores_rating  
FROM airbnb a  
JOIN neighbourhood_avg n ON a.neighbourhood = n.neighbourhood  
WHERE a.price < n.avg_price * 0.75  
      AND a.review_scores_rating >= 4.5  
      AND a.number_of_reviews >= 10  
ORDER BY a.review_scores_rating DESC, a.price ASC;
```

This CTE-based query joins each listing against its neighbourhood average price, then filters for listings that are both highly rated and significantly cheaper than their area average.

Conclusions & Recommendations

1. Location is the strongest pricing driver. Kensington & Chelsea commands £274 avg vs £136 in Greenwich — nearly double for the same room type.
2. Room type matters most after location. Entire homes earn 85% more than private rooms. Hosts should consider converting spare rooms to full listings where possible.
3. Bedrooms add consistent value. Each bedroom adds roughly £16 per night. Larger properties are significantly more profitable per listing.
4. Superhost status does not drive pricing premium. Hosts should focus on location and property type upgrades rather than relying on the Superhost badge to justify higher rates.

5. 459 undervalued opportunities exist. Budget travellers can find highly-rated properties in premium areas like Westminster and Islington at 25%+ below neighbourhood average prices.